Revenue management · Markov decision processes

Airline seat allocation with multiple nested fare classes

Brumelle & McGill, *Operations Research*, 1993: what the paper proves, and what we could reproduce

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The problem

One cabin, several fares and the cheap seats book first

80€	120€	160€	210€	300€
class 5 leisure, books early	class 4	class 3	class 2	class 1 business, books last

Every cheap seat sold today may be a business seat lost tomorrow. The airline must decide **how many seats to protect** for the classes still to come.

Capacity is fixed, demand is random, and the decision is taken **before** knowing it.

Single flight leg, independent demands, no cancellations

The model

The problem is a finite-horizon Markov decision process

The MDP (exercise 3.9 of the course)

Stage: the fare class that is selling
State: s, the number of seats left
Action: accept u ≤ min(s, X) requests
Reward f · u, **transition** s → s − u

Demands are independent, so the future depends only on s: that is the Markov property.

Bellman equation, solved backwards

$$ER_{k+1}[s] = \mathbb{E}\Big[\max_{u \le \min(s,\, X_{k+1})} \big\{ f_{k+1}\, u + ER_k[s-u] \big\}\Big]$$

$ER_0 \equiv 0$: start with class 1, which books last

$$p_k = \arg\max_q \big\{ ER_k[q] - f_{k+1}\, q \big\}$$

$ER_k[s]$ is the value function; the argmax p_k is the protection level, i.e. the optimal policy.

Course ↔ article value $w_k \leftrightarrow ER_k$ · protection level $y_k \leftrightarrow p_k$ · price $p_k \leftrightarrow$ fare f_k · classes in reverse order

Notation of the article: class 1 is the most expensive and books last, X_1 is its demand

Exercise: numbers the classes by increasing price: same model, reversed indices

Before the paper

Two classes were solved in 1972, several classes were guesswork.

Two classes - Littlewood, 1972

protect a seat while

$$f_2 < f_1 \, \mathbb{P}\left[X_1 \geq \text{seats kept}\right]$$

Exact, one line: it is the Bellman equation with two stages.

n classes - EMSRa, Belobaba 1987

apply that rule to **each pair** of classes, then **add up** the seats

Simple, used by airlines, but nobody had shown whether it was right.

The paper answers three questions: **what is optimal**, **is a fixed rule enough**, and **how wrong is EMSRa**.

Notation of the article: class 1 is the most expensive, X_1 its demand

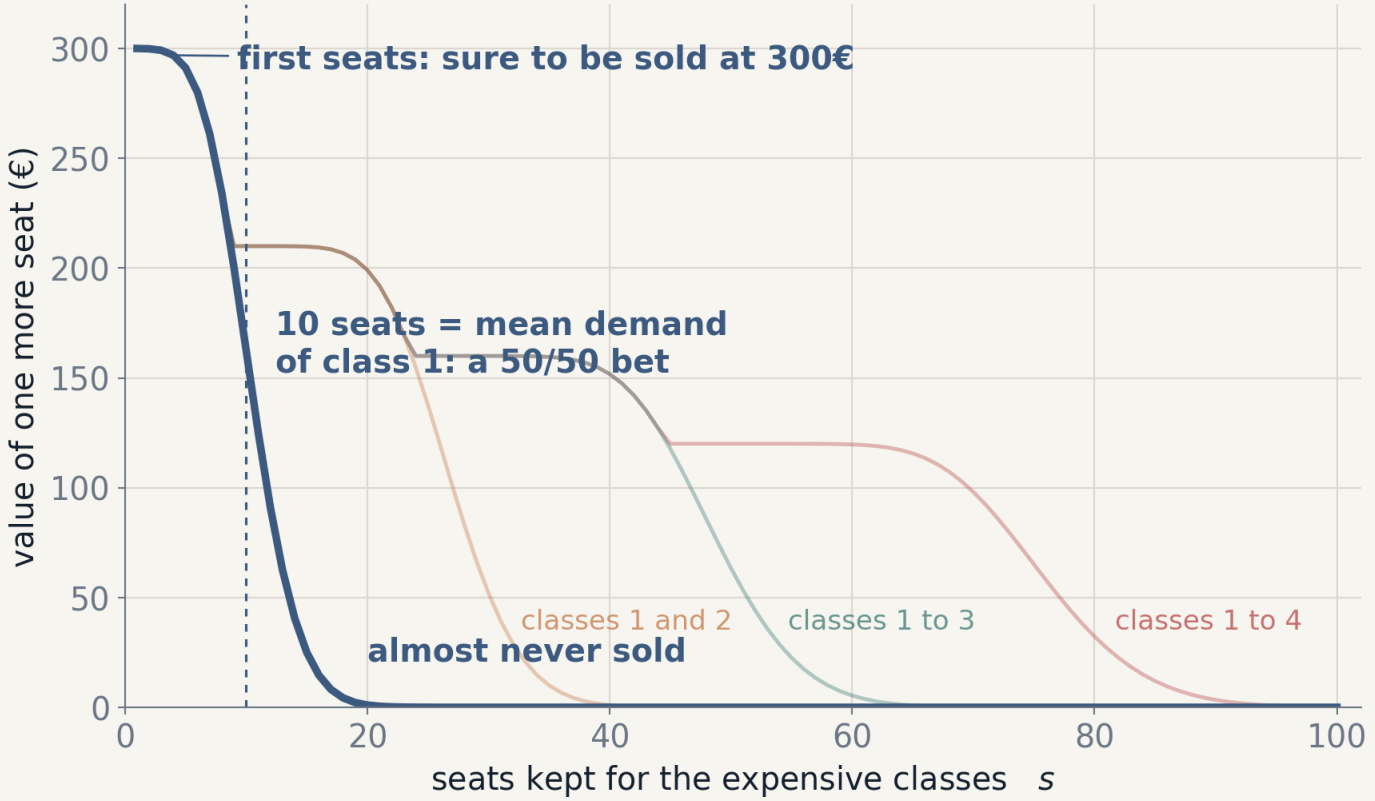
EMSRa: Expected Marginal Seat Revenue, version a

The main result, step 1

How much is one more seat worth?

Class 1	**300€**	10 customers on average
Class 2	**210€**	15 customers
Class 3	**160€**	20 customers
Class 4	**120€**	25 customers
Class 5	**80€**	30 customers, books first

A seat kept for class 1 is worth **300€ × the chance it is bought**. The more seats we keep, the lower that chance.

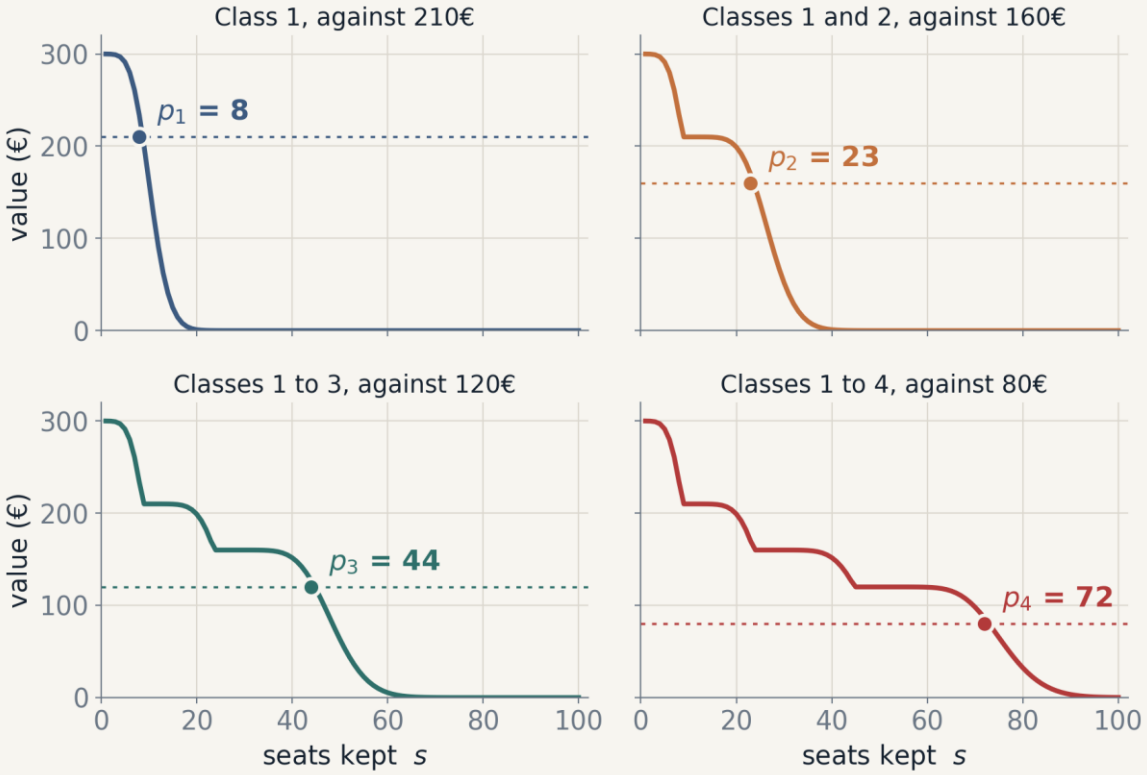


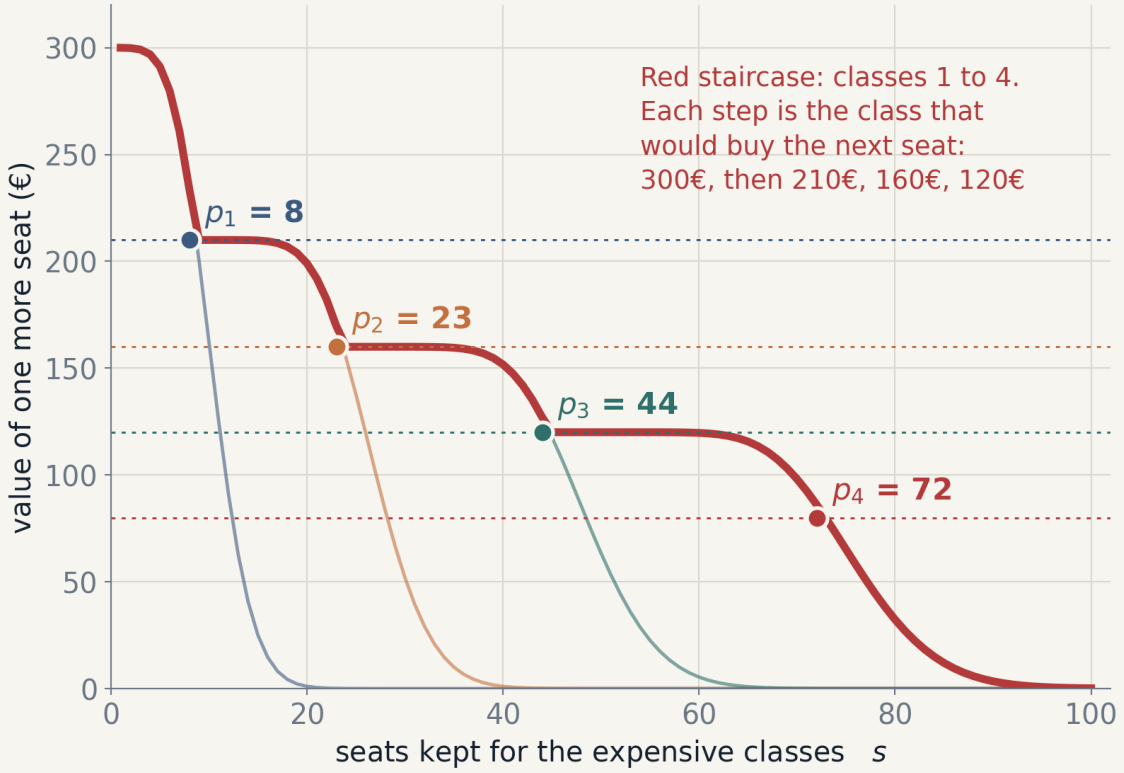
$\delta ER_1[s] = ER_1[s] - ER_1[s-1] = f_1\,\mathbb{P}[X_1 \geq s]$: the slope of the value function, i.e. the bid price of a seat.

100 seats: Poisson demand ; computed with our solver

The main result, step 2

Keep a seat while it is worth more than the fare on the table





$$p_k = \arg\max_q \left\{ ER_k[q] - f_{k+1}\, q \right\} \iff \delta ER_k[p_k + 1] \leq f_{k+1} < \delta ER_k[p_k]$$

The main result, analysis

Protect a seat while it is worth more than the fare on the table

1 · Concave

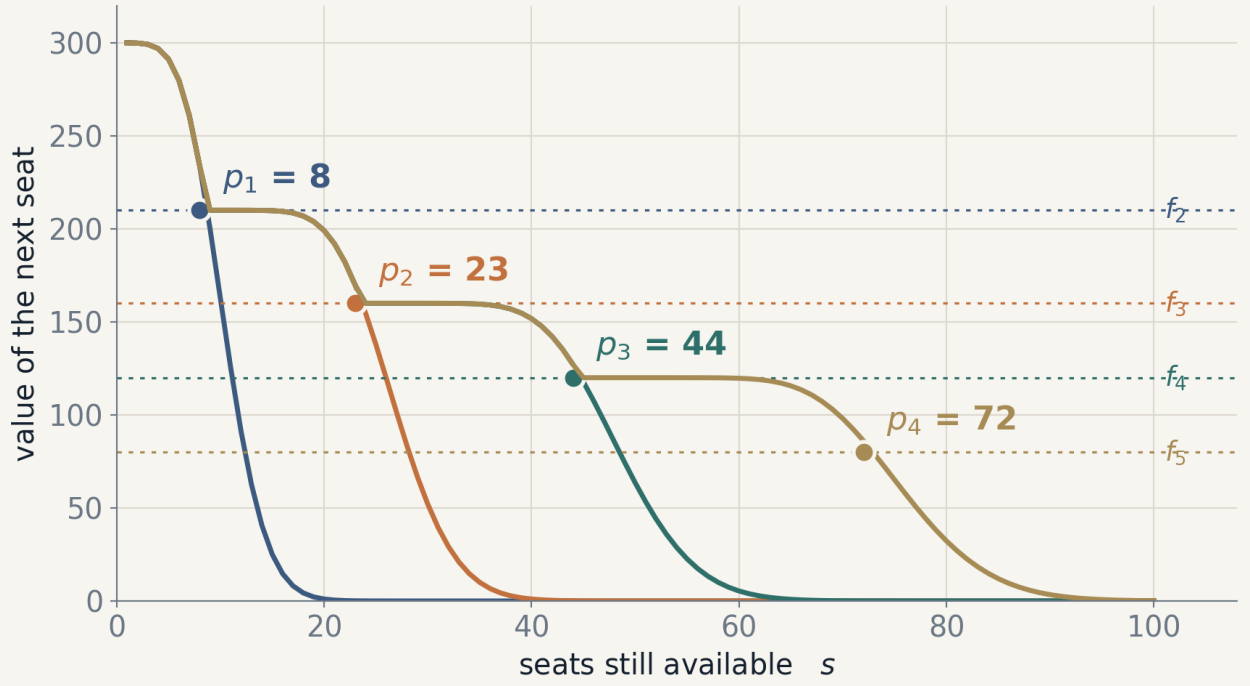
The value function is concave: each extra seat is worth a bit less.

2 · Threshold

So the Bellman argmax is a threshold: sell while more than p_k seats remain.

3 · Nested and optimal

$p_1 \le p_2 \le \dots$ Markov rules suffice: no rule using the booking history does better.



Where the heuristic breaks

A protected seat pays off only if *every* remaining seat is sold

Optimal: one joint probability

$$f_3 = f_1 \, \mathbb{P}\big[X_1 > p_1 \textbf{ and } X_1 + X_2 > p_2\big]$$

The events are linked: the seat is only worth f_1 if the whole cabin fills up.

EMSRa: two separate ones, added up

$$f_3 = f_1 \, \mathbb{P}[X_1 > \pi^1]$$
$$f_3 = f_2 \, \mathbb{P}[X_2 > \pi^2]$$
$$p_2 = \pi^1 + \pi^2$$

Marginals replace the joint law: exact for the top class only.

Both rules agree on the **first level**, but from the second one, EMSRa can be **wrong**, in either direction.

At the optimum, $\delta ER_2[p_2] = f_1 \, \mathbb{P}[X_1 > p_1,\ X_1 + X_2 > p_2]$ (eq. 37), so the rule is $f_3 = \delta ER_2[p_2]$.
EMSRa uses $\mathbb{P}[X_1 > \pi^1]$ and $\mathbb{P}[X_2 > \pi^2]$ separately.

What we implemented

Two algorithms, one answer

Value iteration

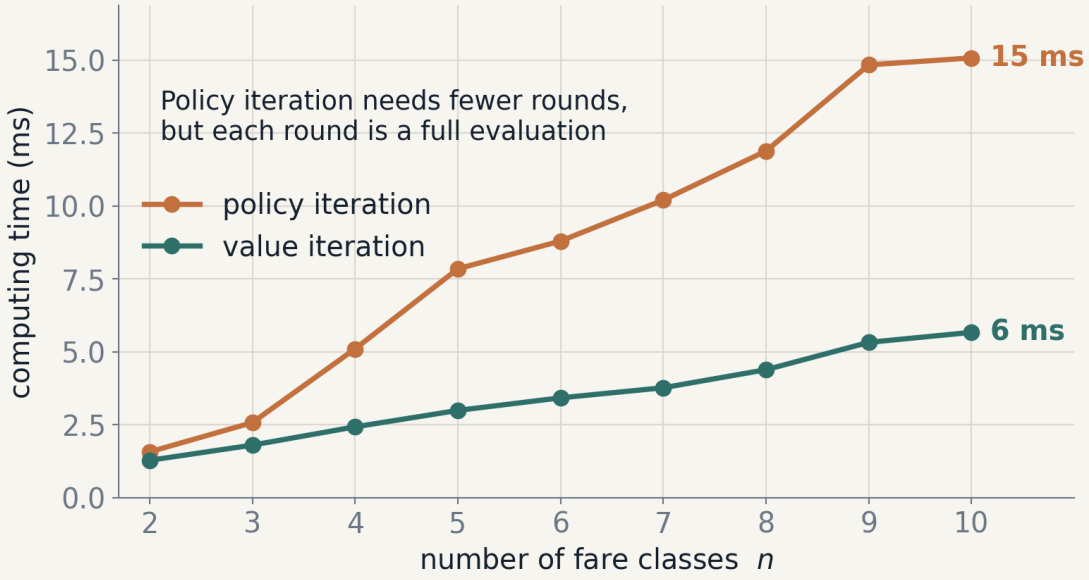
Bellman equation solved backwards over the n stages: one pass, exact.

Policy iteration

Evaluate p, make each p_k greedy, repeat: at most n − 1 useful sweeps, 2 to 4 in practice.

Same answer in 33 runs out of 33

Plus: Littlewood recovered for n = 2, concavity and nesting checked on a 5-class example.



Python + NumPy: a full solve takes under a second for 200 seats

Result 1: validation

Every protection level of the paper, recomputed from scratch

Example	p_1	p_2 optimal	p_2 EMSRa	loss, ours	loss, 1993
$f_2 = 0.7, f_3 = 0.6$	32	80	70	0.33 %	0.37 %
$f_2 = 0.8, f_3 = 0.6$	27	87	80	0.30 %	0.32 %
$f_2 = 0.9, f_3 = 0.6$	19	91	86	0.17 %	0.19 %
$f_2 = 0.8, f_3 = 0.7$	27	75	64	0.36 %	0.41 %
$f_2 = 0.9, f_3 = 0.7$	19	82	73	0.41 %	0.45 %
$f_2 = 0.9, f_3 = 0.8$	19	70	57	0.43 %	0.50 %

Table I of the article: 100 seats, normal demands, $f_1 = 1$ · Table II (capacity effect) reproduced too

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protection levels identical to the published ones

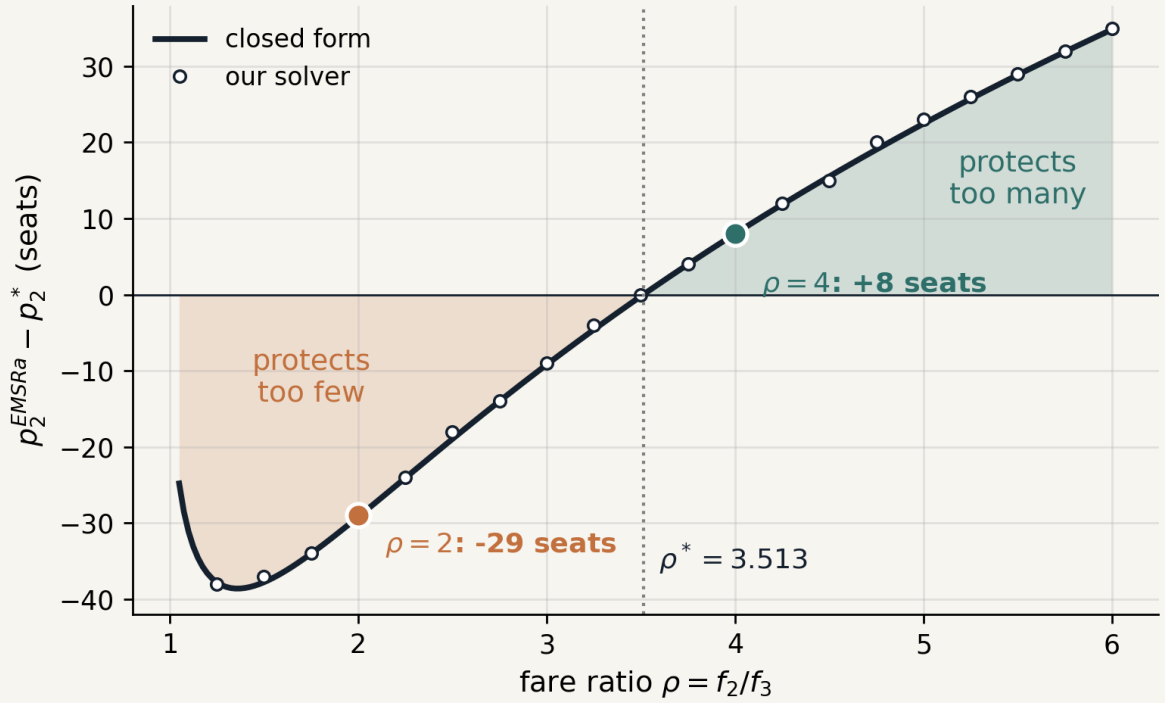
≤ 0.43 %

revenue lost by EMSRa, although it misses the level by 5 to 13 seats

At p^* the slope is zero, so $R(p^* + \Delta) \approx R(p^*) - \frac{1}{2} c \Delta^2$:
doubling the error quadruples the loss.
Ex. 1: $\Delta = 5,\ 10,\ 20 \rightarrow$ loss $0.08,\ 0.33,\ 1.13\,\%$

Result 2 : where EMSRa goes wrong

EMSRa can protect too few or too many seats



Exponential demands, solved by hand

$f_1 = 1,\ \rho = f_2/f_3,\ p_2 = p_1 + \delta,$ unit = 100 seats

Littlewood: $\mathbb{P}[X_1 > p_1] = e^{-p_1} = f_2 \ \Rightarrow\ p_1 = \ln(1/f_2)$

Joint law: $\mathbb{P}[X_1 > p_1,\ X_1 + X_2 > p_2] = (1+\delta)\, e^{-p_2}$

Optimum: $(1+\delta^*)\, e^{-\delta^*} = 1/\rho$

EMSRa: $\pi^1 = \ln(1/f_3),\ \pi^2 = \ln\rho \ \Rightarrow\ \delta^a = 2\ln\rho$

$$\boxed{\delta^a = \delta^* \iff \rho = 1 + 2\ln\rho \ \Rightarrow\ \rho^* \approx 3.513}$$

−29 seats

ρ = 2: protects too few

+8 seats

ρ = 4: protects too many

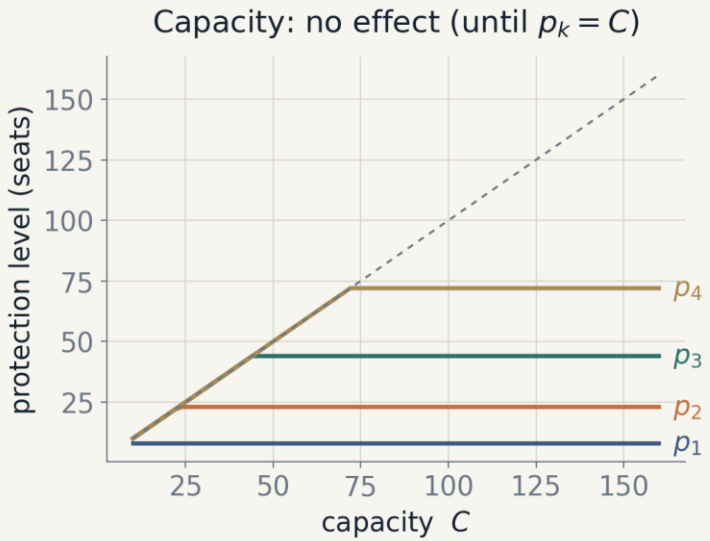
The sign of the error depends on ρ, so no fixed safety margin can correct EMSRa.

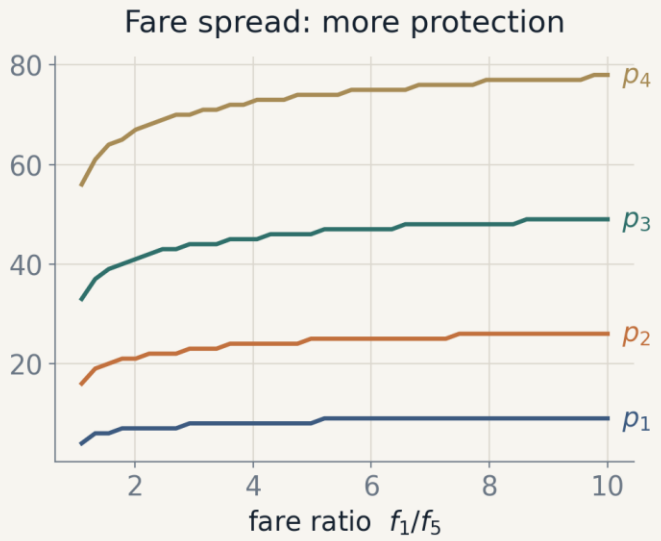
With the normal demands of Table I (ρ ≤ 1.5) it protects 5 to 13 seats too few, yet loses at most 0.43 %.

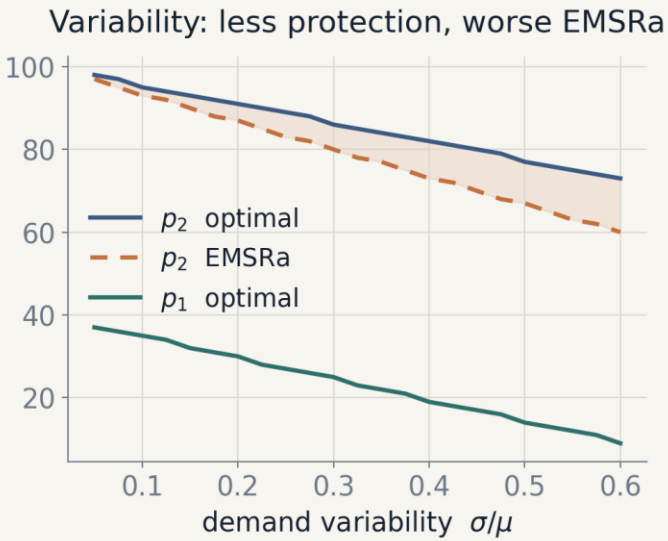
Section 4.1 of the article · line: our closed form · dots: our DP solver and EMSRa on exponential demands (mean 100 seats)

Result 3 : sensitivity

What actually moves the protection levels







Capacity: nothing

C is only the initial state: ER_k does not depend on it, so neither do the levels. They are just capped by C.

Fare spread: a lot

Wider fares, more seats protected, but they stay high even for nearly equal fares.

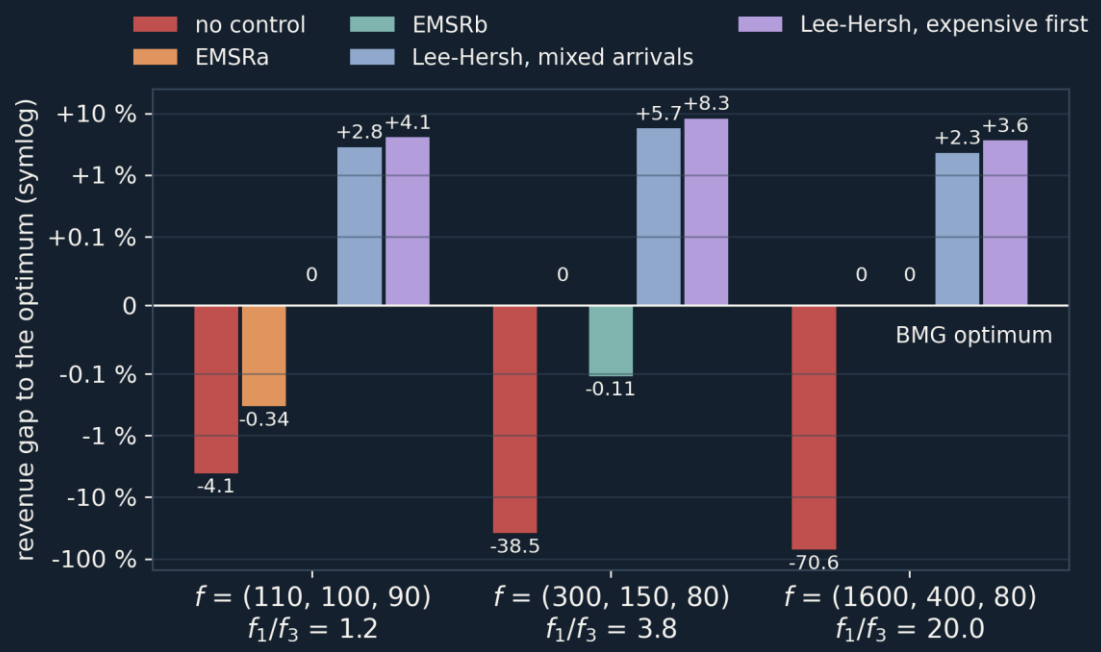
Variability: it decides EMSRa

The gap between EMSRa and the optimum grows from 1 to 13 seats as demand becomes more uncertain.

Our own experiments

Extensions

To go beyond the article



Heuristics stay within 0.4 % of the optimum, no control loses up to 71 %.
Low-before-high is the worst order: otherwise the state is (t, s).

Monte Carlo simulation

$\hat{R} = \frac{1}{N}\sum_i R_i, \qquad \mathrm{SE} = \hat{\sigma}/\sqrt{N}$

$N = 20\,000$: $|\hat{R} - ER_n[C]| = 5.3 < \mathrm{SE} = 7.6$ (gap $0.04\,\%$)

independent check of the exact recursion (9)

EMSRa vs EMSRb

EMSRa: $f_{k+1} = f_j\,\mathbb{P}[X_j > \pi_k^j], \quad p_k = \sum_{j\le k}\pi_k^j$

EMSRb: $f_{k+1} = \bar{f}_k\,\mathbb{P}[X_1 + \cdots + X_k > p_k]$

$\bar{f}_k = \sum_{j\le k} f_j\mu_j \big/ \sum_{j\le k}\mu_j$ (pooled demand: σ grows like $\sqrt{k}$)

Order of arrivals: Lee & Hersh

$V(t,s) = V(t-1,s) + \sum_j \lambda_j(t)\,\big(f_j - \Delta V(t-1,s)\big)^+$

accept $j \iff f_j \ge \Delta V(t-1,s) = V(t-1,s) - V(t-1,s-1)$

low $\to$ high blocks: $V(T,C) = ER_n[C] + 0.03\,\%$, same model

Do you have any questions ?

Graph: 3 classes, Poisson demands (means 10, 15, 20), C = 30, gap to the BMG optimum (symlog scale)